

Database Dictionary

Database Name: iti_project

1- Tables

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Course

Purpose: Stores information about courses offered.

Column	Data Type	Description	Constraints
Crs_id	int	Unique course identifier	PK
name	varchar(50)	Course name	NOT NULL
crs_duration	int	Duration of course	NULL

Department

Purpose: Stores department information.

Column	Data Type	Description	Constraints
Dept_id	int	Unique department ID	PK
Dept_Name	varchar(50)	Department name	NOT NULL
Dept_Location	nvarchar(100)	Department location	NULL
Dept_desc	varchar(250)	Department description	NULL
manger_hiredate	date	Manager hire date	NULL
dept_manger	int	Department manager (Instructor)	FK → Instructor(Ins_id)

Instructor

Purpose: Stores instructor details.

Column	Data Type	Description	Constraints
Ins_id	int	Unique instructor ID	PK
Fname	varchar(50)	First name	NOT NULL
Lname	varchar(50)	Last name	NOT NULL
Salary	decimal(10,2)	Instructor salary	NULL
Dept_id	int	Department ID	FK → Department(Dept_id)

Student

Purpose: Stores student information.

Column	Data Type	Description	Constraints
St_id	int	Unique student ID	PK
address	varchar(100)	Student address	NULL
birthdate	date	Date of birth	NULL
Fname	varchar(50)	First name	NOT NULL
Lname	varchar(50)	Last name	NOT NULL
Dept_id	int	Department ID	FK → Department(Dept_id)
Password	varchar(50)	Student password	NOT NULL

Question

Purpose: Stores questions for exams.

Column	Data Type	Description	Constraints
Q_id	int	Unique question ID	PK
Q_type	varchar(10)	Question type (mcq / t&f)	CHECK constraint
mark	int	Marks assigned to question	NULL
crs_id	int	Course ID	FK → Course(Crs_id)
Q_text	varchar(500)	Question text	NULL

Choice

Purpose: Stores choices for MCQ questions.

Column	Data Type	Description	Constraints
Ch_id	int	Unique choice ID	PK
text	varchar(500)	Choice text	NULL
is_correct	bit	Indicates if choice is correct	NULL
Q_id	int	Question ID	FK → Question(Q_id)

Exam

Purpose: Stores exam information.

Column	Data Type	Description	Constraints
exam_id	int	Unique exam ID	PK
exam_name	varchar(100)	Exam name	NOT NULL
total_marks	int	Total exam marks	NULL
Crs_id	int	Course ID	FK → Course(Crs_id)

ExamQuestions

Purpose: Links exams with questions (Many-to-Many).

Column	Data Type	Description	Constraints
exam_id	int	Exam ID	PK, FK → Exam
Q_id	int	Question ID	PK, FK → Question

Stud_Course

Purpose: Links students with courses.

Column	Data Type	Description	Constraints
St_id	int	Student ID	PK, FK → Student
Crs_id	int	Course ID	PK, FK → Course
start_date	date	Course start date	NULL
finish_date	date	Course finish date	NULL
grade	decimal(5,2)	Student course grade	NULL

Ins_Course

Purpose: Links instructors with courses.

Column	Data Type	Description	Constraints
Ins_id	int	Instructor ID	PK, FK → Instructor
Crs_id	int	Course ID	PK, FK → Course
evaluation	int	Instructor evaluation (0-100)	CHECK (0-100)

StudentExam

Purpose: Stores student exam results.

Column	Data Type	Description	Constraints
st_id	int	Student ID	PK, FK → Student
exam_id	int	Exam ID	PK, FK → Exam
grade	int	Exam grade (0-100)	CHECK (0-100)

StudentAnswers

Purpose: Stores student answers per exam question.

Column	Data Type	Description	Constraints
st_id	int	Student ID	PK, FK → Student
exam_id	int	Exam ID	PK, FK → Exam
Q_id	int	Question ID	PK, FK → Question
Choise_id	int	Selected choice	FK → Choise(Ch_id)

Topic

Purpose: Stores course topics.

Column	Data Type	Description	Constraints
topic_id	int	Topic ID	PK
crs_id	int	Course ID	PK, FK → Course
topic_name	varchar(100)	Topic name	PK

2- Stored Procedures

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Exam_Correction

Corrects a student exam and calculates final grade.

Parameters: @St_id INT, @Exam_id INT

exam_generation

Generates a random exam.

Parameters: @Crs_id INT, @MCQ_count INT, @TF_count INT,
@Exam_id INT OUTPUT

GetStudentGrades

Returns grades of a student.

Parameters: @StudentId INT

GetInstructorCoursesWithStudentCount

Returns instructor courses with student count.

Parameters: @InstructorId INT

GetCourseTopics

Returns topics for a course.

Parameters: @CourseId INT

GetExamQuestionsWithChoices

Returns exam questions with choices.

Parameters: @ExamId INT

GetExamQuestionsWithStudentAnswers

Returns exam questions with student answers.

Parameters: @ExamId INT, @StudentId INT

GetRandomExamByCourse

Returns random exam with its Questions and checks that the student didn't take it before.

Parameters: @Crs_id INT, @St_id Int

sp_LoginStudent

Handles student authentication by validating the student ID and password against the Student table.

Parameters: @St_id INT,@Password VARCHAR(50)

Submit_Exam_Answers

Allows a student to submit answers for an exam.

Parameters: @St_id INT,@Exam_id INT,@Choice_IDs VARCHAR(255).

rpt_StudentsByDepartment

Returns a report listing all students belonging to a specific department.

Parameters: @DeptID INT.

CRUD Stored Procedures

Insert Procedures

- SP_Insert_Student (@St_id INT , @address varchar(100),@birthdate Date, @Fname varchar(50), @Lname varchar(50), @Dept_id INT)
- SP_Insert_Instructor (@Ins_id INT,@Fname varchar(50),@Lname varchar(50),@Salary decimal(10,2),@Dept_id INT)
- SP_Insert_Department (@Dept_id INT, @Dept_name varchar(50),@Dept_Location nvarchar(100),Dept_desc varchar(250),@manger_hiredDate Date, @dept_manger Int)
- SP_Insert_Course (@Crs_id INT, @name varchar(50),@crs_duration INT)
- SP_Insert_Topic (@topic_id INT , @crs_id INT, @topic_name varchar(100))
- SP_Insert_Question (@Q_id INT , Q_text varchar(500), Q_type varchar(10),@mark INT, @Crs_id INT)
- SP_Insert_Choice (@Ch_id INT , @text varchar(500),@is_correct bit, @Q_id INT)
- SP_Insert_Exam (@exam_id INT,exam_name varchar(100),@total_marks INT, @Crs_id Int)
- SP_Insert_ExamQuestions (@exam_id INT, @Q_id INT)
- SP_Insert_Ins_Course (@ins_id INT, @Crs_id Int, @evaluation INT)

- SP_Insert_Stud_Course (@St_id INT, @Crs_id INT, @start_date Date,@finish_date Date, @grade decimal(5,2))
- SP_Insert_StudentAnswers (@st_id INT, @exam_id INT, @Choise_id INT, Q_id INT)
- SP_Insert_StudentExam (@st_id INT, @exam_id INT , @grade INT)

Update Procedures

- SP_Update_Student (@St_id INT , @address varchar(100),@birthdate Date, @Fname varchar(50), @Lname varchar(50), @Dept_id INT)
- SP_Update_Instructor (@Ins_id INT,@Fname varchar(50),@Lname varchar(50),@Salary decimal(10,2),@Dept_id INT)
- SP_Update_Department (@Dept_id INT, @Dept_name varchar(50),@Dept_Location nvarchar(100),Dept_desc varchar(250),@manger_hiredat Date, @dept_manger Int)
- SP_Update_Course (@Crs_id INT, @name varchar(50),@crs_duration INT)
- SP_Update_Topic (@topic_id INT , @crs_id INT, @old_topic_name varchar(100),@new_topic_name)
- SP_Update_Question (@Q_id INT , Q_text varchar(500), Q_type varchar(10),@mark INT, @Crs_id INT)
- SP_Update_Choice (@Ch_id INT , @text varchar(500),@is_correct bit, @Q_id INT)
- SP_Update_Exam (@exam_id INT,exam_name varchar(100),@total_marks INT, @Crs_id Int)
- SP_Update_ExamQuestions (@old_exam_id INT,@old_Q_id INT,@new_exam_id INT, @new_Q_id INT)
- SP_Update_Ins_Course (@ins_id INT, @Crs_id Int, @evaluation INT)
- SP_Update_Stud_Course (@St_id INT, @Crs_id INT, @start_date Date,@finish_date Date, @grade decimal(5,2))
- SP_Update_StudentAnswers (@st_id INT, @exam_id INT, Q_id INT, @new_Choise_id INT,)
- SP_Update_StudentExam (@st_id INT, @exam_id INT , @grade INT)

Delete Procedures

- SP_Delete_Student (@St_id INT)
- SP_Delete_Instructor(@Ins_id INT)

- SP_Delete_Department (@Dept_id INT)
- SP_Delete_Course (@Crs_id INT)
- SP_Delete_Topic (@topic_id INT , @crs_id INT, @topic_name varchar(50))
- SP_Delete_Question (@Q_id INT)
- SP_Delete_Choice (@Ch_id INT)
- SP_Delete_Exam (@exam_id INT)
- SP_Delete_ExamQuestions (@exam_id INT, @Q_id INT)
- SP_Delete_Ins_Course (@Ins_id INT, @Crs_id INT)
- SP_Delete_Stud_Course (@St_id INT, @Crs_id INT)
- SP_Delete_StudentAnswers (@st_id INT , @exam_id INT , @Choise_id INT, Q_id INT)
- SP_Delete_StudentExam (@st_id INT , @ exam_id INT)

Select Procedures

The Parameters in Select is optional only for filtering

- SP_Select_Student (@St_id INT)
- SP_Select_Instructor (@Ins_id INT)
- SP_Select_Department (@Dept_id INT)
- SP_Select_Course (@Crs_id INT)
- SP_Select_Topic (@topic_id INT, crs_id INT)
- SP_Select_Question (@Q_id)
- SP_Select_Choice (@Ch_id INT, @Q_id INT)
- SP_Select_Exam (@exam_id INT)
- SP_Select_ExamQuestions (@exam_id INT , Q_id INT)
- SP_Select_Ins_Course (@Ins_id INT, @Crs_id INT)
- SP_Select_Stud_Course (@St_id INT , @Crs_id INT)
- SP_Select_StudentAnswers (@St_id INT , @exam_id INT, Q_id INT)
- SP_Select_StudentExam (@st_id Int, @exam_id INT)

